

Same Formula, Different Signal: Funding-Rate Censoring and Regime Dynamics in Decentralised Perpetual Futures

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Abstract

Hyperliquid, Binance and Bybit set perpetual funding by the same rule, which pays exactly an interest floor of 0.01% per eight hours whenever the premium lies inside a 10 bp band. Over 2 October 2025 to 28 April 2026 the same formula sends different signals: 46.0% of Hyperliquid's 5,001 hourly settlements sit exactly on the floor, and 48.4% at the exchanges' settlement hours, against 5.9% on Binance and 7.7% on Bybit. The cause is where the premiums sit — the exchanges' just below the band, Hyperliquid's across its lower edge with a long tail into it — and the gap persists over the five months that follow, 51.7% against 10.9% and 7.9%. Hyperliquid's headline rate is censored about half the time; its premium, not its rate, carries the signal.

Fitted to that premium, a two-state Markov-switching AR(1) reverts in 36 hours when calm and 3.2 under stress, a ratio of 11 with a 95% interval of [3, 19]. One hour carries its significance. Replacing the crash hour of 10 October 2025 with the next largest observation leaves 3.0 on an interval covering unity, only one of fourteen rolling windows lies significantly above unity, and the five months after the sample reverse the ordering, at 0.23, its 95% interval ending at 0.49. The premium likelihood also has a dominant basin that is not its maximum: 24 of 40 estimations settle there and report a ratio of 25. On returns, a three-state Gaussian HMM separates hourly volatility into 14.6%, 33.3% and 88.4% annualised states, and the estimated probability of a direct exit from the high state to the low is zero, with a 95% bound of 0.3% per hour. All results are descriptive.

Keywords: funding rates; censoring; perpetual futures; regime switching; hidden Markov models; Hyperliquid; Binance; Bybit; Hamilton filter.

JEL: C22, C58, G12, G13.

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1 Introduction

Perpetual futures dominate cryptocurrency derivatives, with daily volumes above \$100 billion in 2022, two to three times spot turnover [17], and decentralised venues, Hyperliquid foremost, have become structurally significant. Having no expiry, a perpetual is tied to its index by a recurring funding payment between longs and shorts, set from an averaged premium [1, 19]. Hyperliquid, Binance and Bybit compute it with the same rule, and inside a band of premia that rule pays exactly the interest rate, so the published rate stops moving with the market (Figure 2a). How often that happens decides whether the headline rate can be read as a signal at all.

Funding has a literature on the carry and basis side. Schmeling et al. [29] read crypto carry as compensation for the limits to arbitrage that anchor perpetuals to spot, Makarov and Schoar [27] document the frictions segmenting these markets, Alexander et al. [2] study price discovery on the first large perpetual venue, Werapun et al. [31] compare funding arbitrage across centralised and decentralised exchanges, and Hugonnier and Jermann [18] model perpetual pricing with the clamp explicitly and report a bimodal distribution of perpetual prices on Binance that they attribute, tentatively, to it. Regime detection in Bitcoin returns is well studied [13, 24, 26]: volatility clusters and breaks [4], tails are heavy [24] and activity moves with returns [25], which the Markov-switching framework of Hamilton [15] and the hidden Markov machinery of Rabiner [28] capture with a few latent states. Zhivkov et al. [33] apply Bai–Perron tests to rolling cross-exchange integration statistics across 26 venues and find no discrete structural breaks, and a companion paper documents a two-tiered funding market [32]; their object is an integration statistic tested for breaks, ours a latent-state decomposition of one venue, but a reader should know that a study on overlapping data finds no breaks where we find states.

We make three contributions, all descriptive and in-sample. First, a measurement: one rule pins the headline rate to the floor in 46.0% of Hyperliquid’s hours and in 5.9% and 7.7% of Binance’s and Bybit’s settlements, because the exchanges’ premiums sit just below the band while Hyperliquid’s straddles its edge (Section 3.2). Headline funding compared across venues therefore mixes censoring with differences in premium, and on Hyperliquid the premium is the object to model. Second, a joint regime description of the return and premium processes of one decentralised perpetual, whereas the regime literature rests almost entirely on spot prices at centralised venues: three hourly volatility states whose direct move from high to low is estimated at zero, on the boundary, and bounded below 0.3% per hour (Section 4), and a premium whose apparent asymmetry in mean reversion belongs to a single episode, the crash hour of 10 October 2025 (Section 5). Third, a caution that travels: the premium likelihood is multi-modal, and its most frequent optimum, which a single estimation reports reliably enough to look converged, is not its maximum (Figure 7). Figure 1 maps the windows and data behind every result.

2 Econometric Framework

2.1 Gaussian Hidden Markov Model on Returns

Let r_t be the hourly log-return and $S_t \in \{1, \dots, K\}$ a latent state. The Gaussian HMM has the joint law

$$\mathbb{P}(r_{1:T}, S_{1:T} \mid \theta) = \pi_{S_1} b_{S_1}(r_1) \prod_{t=2}^T A_{S_{t-1}, S_t} b_{S_t}(r_t), \quad (1)$$

with $\pi_i = \mathbb{P}(S_1 = i)$, $A_{ij} = \mathbb{P}(S_t = j \mid S_{t-1} = i)$ and emission $b_k(r) = \varphi(r; \mu_k, \sigma_k^2)$. The *filtered* posterior $\mathbb{P}(S_t = k \mid \mathcal{F}_t)$, with $\mathcal{F}_t = \sigma(r_{1:t})$, uses only the past and is the one admissible for real-time use; the *smoothed* posterior $\mathbb{P}(S_t = k \mid \mathcal{F}_T)$ conditions on the whole sample and serves only as an in-sample diagnostic. We estimate by Baum–Welch from eight random starts, keep the highest likelihood, and order states by $\hat{\sigma}_k$ ascending — low, moderate and high volatility — because the means are too close to order by; Appendix B gives the recursions and the case against a μ -ascending rule.

Model selection uses BIC over $K \in \{2, 3\}$, with $K^2 + 2K - 1$ free parameters, 7 and 14; Section 4.3 adds $K=4$ and 5. A likelihood-ratio test is not available: under $K=2$ the third state’s parameters are unidentified and the statistic has no χ^2 limit [7, 9, 16], so we read the criteria as evidence, not as tests at a nominal size. Penalised-likelihood order estimation is consistent for mixtures and for finite-alphabet hidden Markov models [12, 21].

2.2 Markov-Switching AR(1) for the Funding Premium

For the premium y_t in basis points we use the mean-adjusted two-state form of Hamilton [15],

$$y_t - \mu_{S_t} = \phi_{S_t}(y_{t-1} - \mu_{S_{t-1}}) + \sigma_{S_t} \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad (2)$$

in which μ_k is the regime *level*, not an intercept, all of $(\mu_k, \phi_k, \sigma_k)$ switch, and the likelihood conditions on the pair (S_t, S_{t-1}) . The half-life

$$\tau_k = \frac{\ln 0.5}{\ln |\phi_k|} \quad (3)$$

says in hours how long half of a premium dislocation takes to decay within regime k . We estimate by maximum likelihood with the Hamilton filter and the smoother of Kim [22] as implemented in **statsmodels** and, because the likelihood is multi-modal (Section 5.3), always from 40 estimations of 40 random initialisations each, reporting the highest terminal log-likelihood.

2.3 Conditional Moments

With the predicted state distribution $\tilde{p}_{j,t} \equiv \mathbb{P}(S_{t+1} = j \mid \mathcal{F}_t) = \sum_k \mathbb{P}(S_t = k \mid \mathcal{F}_t) A_{kj}$, the one-step conditional mean and variance of returns are

$$\mathbb{E}[r_{t+1} \mid \mathcal{F}_t] = \sum_{j=1}^K \tilde{p}_{j,t} \mu_j \equiv \bar{\mu}_t, \quad (4)$$

$$\text{Var}(r_{t+1} \mid \mathcal{F}_t) = \underbrace{\sum_j \tilde{p}_{j,t} \sigma_j^2}_{\text{within-regime}} + \underbrace{\sum_j \tilde{p}_{j,t} (\mu_j - \bar{\mu}_t)^2}_{\text{between-regime}}, \quad (5)$$

the second term being the variance that regime uncertainty itself adds, a feature absent from single-regime stochastic-volatility models.

3 Data and Funding Mechanics

3.1 Sample Construction

For the BTC perpetual we take Hyperliquid’s hourly candles of traded prices and its hourly funding observations from the public endpoints **candleSnapshot** and **fundingHistory**, floored to the hour and inner-joined: $T = 5,001$ observations from 2 October 2025 00:00 UTC to 28 April 2026 08:00 UTC, retrieved on 28 April 2026. For Section 3.2 we add, retrieved on 6 October 2026, the funding histories and hourly premium index of the Binance and Bybit BTCUSDT perpetuals from 1 October 2025 to 30 September 2026. Beyond the estimation window, every funding series runs to the settlement stamped 1 October 2026 00:00 UTC, which pays for the final hours of 30 September. Figure 1 shows the windows and which series survive at which resolution.

The data set the window. **fundingHistory** keeps a contract’s whole history, and returns an unbroken hourly BTC series from June 2023; **candleSnapshot** serves only the most recent 5,000 or so candles at each interval, so it reaches some 208 days back at hourly resolution, 417 at two-hourly and 833 at four-hourly. The premium series is therefore replicable and extensible in both directions; the hourly return series is not, its first two months having left the hourly window before the panel was archived. Two-hourly candles re-retrieved in August 2026 agree bit for bit with the surviving hourly panel aggregated to two hours on all 1,753 overlapping bars; they bound the return extrema of Table 1 and check the state structure in Section 4.3. Both models use the common 5,001-hour window, which contains the local high of \$126,101 and the February 2026 trough of \$62,883.

Returns are $r_t = \ln(p_t/p_{t-1})$ on the last traded price of hour t , not the mark price, in basis points; funding is in basis points per hour and the premium in 8-hour-equivalent basis points. Annualisation uses $\sqrt{24 \cdot 365}$ for volatility and $24 \cdot 365$ for funding.

3.2 The Funding Formula and the Clamp Artifact

Hyperliquid’s documentation specifies the 8-hour-equivalent funding rate as

$$F_t = \bar{P}_t + \text{clamp}(r - \bar{P}_t, -5 \cdot 10^{-4}, +5 \cdot 10^{-4}), \quad (6)$$

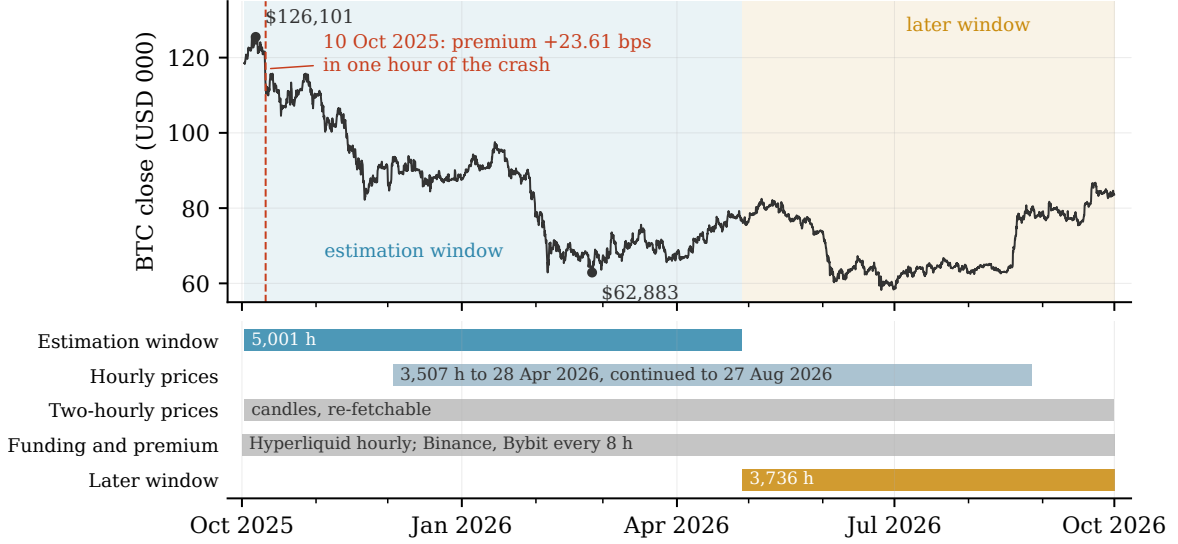


Figure 1. Windows and data. **Top:** BTC two-hourly close, the estimation window (2 October 2025 – 28 April 2026) and the later window (28 April – 30 September 2026) shaded, with the price extremes of Table 1 and the premium’s maximum hour marked. **Bottom:** the windows and the series the replication package ships; hourly prices survive only from 3 December 2025.

Table 1. Summary statistics for hourly observations on Hyperliquid BTC perpetual, 2 Oct 2025 – 28 Apr 2026 ($T = 5,001$).

Series	Mean	Std. dev.	Min	Median	Max	Skew
Log-return (bps/h)	−0.87	50.92	†	−0.27	†	−0.21
Funding rate (bps/h)	0.059	0.103	−0.694	0.111	2.327	+0.94
Premium (bps)	−3.15	2.87	−10.55	−4.11	23.61	+1.71
Closing price (USD)	86,314	15,988	62,883	87,678	126,101	—

† Withheld. The values recorded at retrieval, $-3,800$ and $+3,421$ bps/h, are inconsistent with the standard deviation beside them: the two alone would contribute twice the sample’s sum of squared deviations. Mean and standard deviation are reproduced by the fitted mixture of Table 4, -0.87 and 50.95 bps/h, so the extrema are in error (median and skewness of a dependent sample need not match the mixture’s). The two-hourly candles bound every hourly move by the high–low range of its two-hour bar, at most $1,498$ bps (14.98% , on 10 October 2025), excluding the recorded values by factors of 2.3 and 2.5 . On the hourly sub-period of Section 4.3 the realised range is $[-367.5, +336.7]$ bps/h.

with $\bar{P}_t$ the premium index averaged over the preceding hour, $r = 10^{-4}$ the interest rate per eight hours and $\text{clamp}(x, a, b) = \max(a, \min(b, x))$; the hourly payment is $F_t/8$ [19]. Binance and Bybit use the same 0.01% interest and $\pm 0.05\%$ clamp [5, 6], with which Hyperliquid states its rule is designed to be consistent; Kim and Park [23] study the design problem such a rule solves. The venues differ in schedule — hourly against every eight hours — and in how each builds its premium index, from its own index price and impact size, and averages it over its own settlement interval.

Figure 2(a) draws the rule. Whenever $\bar{P}_t \in (-4, +6)$ bps — a band centred on $+1$ bp, not on zero — the clamp is the identity and $F_t = r$: the hourly rate is exactly 0.125 bps/h, which we call the interest floor, although rates fall below it whenever the premium lies below the band. Hyperliquid’s premium has a mean of -3.15 bps, just inside the band’s lower edge, and $2,300$ of $5,001$ hours — 46.0% — sit exactly on the floor; the count is exact, the floor hours being precisely those with $\bar{P}_t$ inside the band. The hourly schedule does not cause the share. Read every eighth hour, the series gives 44.2% to 48.4% by phase offset, averaging 46.0% ; the exchanges’ 00, 08 and 16 UTC stamps give the top of that range, but only one point above the hourly share after the window (52.7% against 51.7% , Table 2), differences of a few points beside a gap of forty. An eight-hour average, with a standard deviation of 2.75 bps against 2.87 , even lands in the band more often, 48.0% of the time: the schedule multiplies censored payments, not censored observations.

The point mass breaks any Gaussian regime model of F_t : an MS-AR fitted to the rate returns a degenerate regime with $\hat{\sigma}^2$ of order 10^{-32} pinned to the floor and one ordinary regime with $\hat{\phi} = 0.81$. We therefore model the premium $\bar{P}_t$, which `fundingHistory` publishes beside each rate. The two series point in opposite directions (Table 1): funding averaged $+0.059$ bps/h, about $+5.2\%$ a year, so longs

paid shorts, while the premium averaged -3.15 bps, the contract trading below its index for much of the sample. In the 46.0% of hours with the premium inside the band the contract paid the floor whatever the positioning, and a reader of the headline rate as positioning would see longs paying over a window in which the contract mostly traded below its index.

Same formula, different signal. Binance and Bybit settle every eight hours under the same rule, and their funding histories make the comparison exact: a settlement equals the floor exactly when the averaged premium sat inside the band, and any other settlement identifies that premium, as the settlement minus 0.05% below the band or plus 0.05% above it. Over the paper window Binance prints the floor in 37 of 626 settlements (5.9%) and Bybit in 48 (7.7%), against 46.0% on Hyperliquid, which still prints it 48.4% of the time read at the exchanges' 00, 08 and 16 UTC stamps (Figure 2c, Table 2). No exchange settlement exceeds the floor; 94.1% of Binance's and 92.3% of Bybit's lie below it. The five months after the window repeat the pattern, 51.7% on Hyperliquid against 10.9% and 7.9%.

The premiums explain the difference (Figure 2b). Recovered from their settlements, the exchanges' premiums are concentrated just below the band, with medians of -4.75 and -4.82 bps and interquartile ranges of 0.66 and 0.60 bp. Hyperliquid's straddles the edge — median -4.11 bps, 52.5% of hours below the band — with a long tail into the band that lifts its mean to -3.15 . The censored share is a property of where a venue's premium sits, not of the formula, which is shared, nor of when settlements are sampled, although how each venue averages its premium may contribute: an eight-hour average narrows a distribution whose mean lies below the band and so lowers the share inside it. Funding-arbitrage studies rightly use the cash flow [31]; read as positioning, headline funding compared across venues sets a series censored half the time against ones that rarely are, and the premiums, which the rule does not censor, are the better basis, though their windows and indices differ.

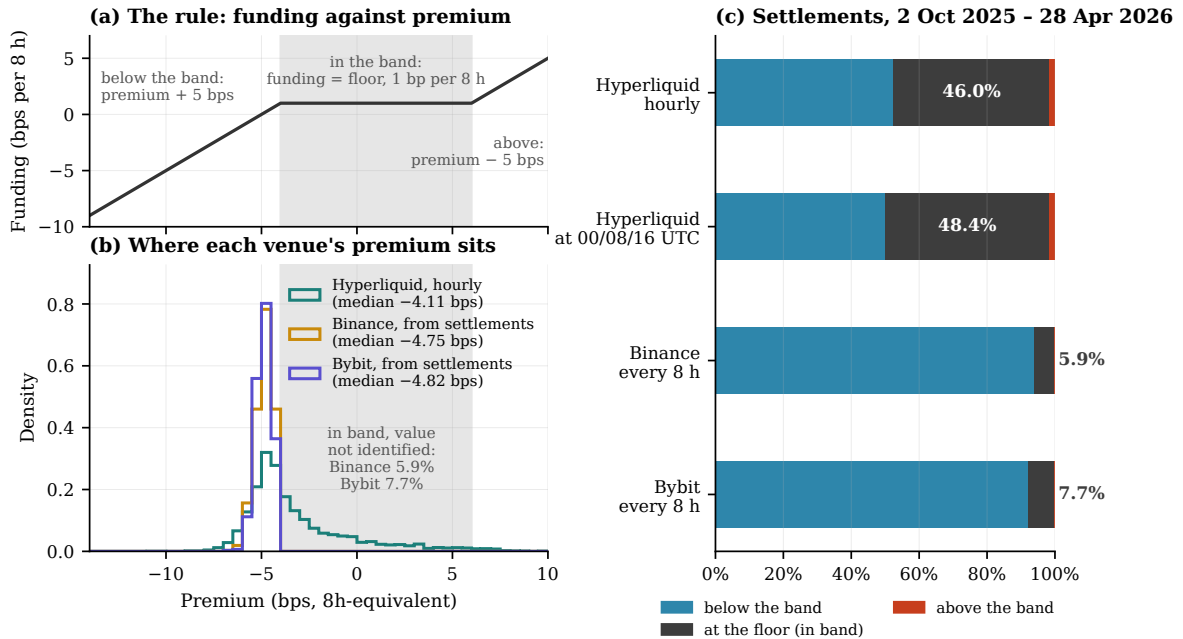


Figure 2. Funding-rate censoring across venues, BTC perptuals, 2 October 2025 – 28 April 2026. **(a)** The rule (6): funding per eight hours against the averaged premium; inside the band ($-4, +6$) bps it pays exactly the floor of 1 bp. **(b)** Where each venue's premium sits: Hyperliquid's hourly average premium, and the Binance and Bybit premiums recovered exactly from their settlements. Settlements at the floor say only that the premium lay inside the band and are not drawn, so each exchange's curve integrates to its identified share. **(c)** Shares of settlements below the band, exactly at the floor, and above it: Hyperliquid hourly and read at 00, 08 and 16 UTC, and the eight-hourly settlements of Binance and Bybit.

Table 2. Funding settlements exactly at the interest floor, compared as decimals, per venue and window. *Below*: settlements under the floor, the premium below the band. *Median* and *IQR*: premium in bps, Hyperliquid’s hourly average and the exchanges’ recovered from settlements; floor settlements rank between those below and above the band, so the quantiles are exact. The second Hyperliquid row reads the hourly series at the exchanges’ settlement stamps.

Venue	Settles	2 Oct 2025 – 28 Apr 2026					28 Apr – 30 Sep 2026	
		n	At floor	Below	Median	IQR	n	At floor
Hyperliquid	hourly	5,001	46.0%	52.5%	−4.11	[−4.93, −2.26]	3,736	51.7%
Hyperliquid	at 00/08/16 UTC	626	48.4%	50.0%			467	52.7%
Binance	every 8 h	626	5.9%	94.1%	−4.75	[−5.10, −4.44]	467	10.9%
Bybit	every 8 h	626	7.7%	92.3%	−4.82	[−5.09, −4.49]	467	7.9%

4 Regime Estimation: Returns

4.1 Model Selection

BIC prefers $K=3$ to $K=2$ by about 217 points (Table 3): the third state’s likelihood gain outweighs its seven extra parameters. We read the difference as evidence, not as a test at a nominal size, and attach no Bayes-factor reading to its magnitude [20].

Table 3. Information criteria for $K \in \{2, 3\}$ on $T = 5,001$ hourly log-returns.

K	$\log \mathcal{L}$	# params	AIC	BIC
2	20,132.0	7	−40,250.0	−40,204.4
3	20,270.2	14	−40,512.4	−40,421.1

4.2 Regime Parameters and Geometry

Table 4 reports the $K=3$ fit. The states separate on volatility and only on volatility: 15.6, 35.6 and 94.4 bps/h, or 14.6%, 33.3% and 88.4% annualised, with bootstrap intervals of [13.6, 15.5], [31.8, 34.9] and [83.0, 94.0] per cent that are nowhere near overlapping, and stationary masses of 20, 60 and 20%. No drift separates from zero: the largest, $\hat{\mu}_3 = -5.69$ bps/h, has a standard error of 3.28 and an interval of [−12.44, +0.27], so we do not call the high-volatility state one of negative expected return.

Table 4. Regime parameters of the $K=3$ Gaussian HMM on hourly log-returns. Annualised σ uses $\sigma\sqrt{24 \cdot 365}$; expected duration is $\mathbb{E}[\tau_k] = 1/(1 - A_{kk})$. Parenthesised figures are parametric-bootstrap standard errors (Section 4.4).

State	$\hat{\mu}_k$ (bps/h)	$\hat{\sigma}_k$ (bps/h)	Annualised $\hat{\sigma}_k$	$\hat{A}_{kk}$	$\mathbb{E}[\tau_k]$ (h)
1 low	0.139 (0.62)	15.55 (0.52)	14.6% (0.49)	0.9235 (0.0120)	13.1 (2.08)
2 moderate	0.404 (0.78)	35.57 (0.87)	33.3% (0.82)	0.8867 (0.0114)	8.8 (0.90)
3 high	−5.685 (3.28)	94.42 (2.97)	88.4% (2.78)	0.7307 (0.0303)	3.7 (0.42)

These are hourly states. Expected durations of 13.1, 8.8 and 3.7 hours imply some 686 state visits over 5,001 hours, one switch about every seven, and some 270 entries into the high state in seven months: a three-bucket classifier of hourly conditional volatility, not a description of market phases. The fitted chain, with geometric durations of a few hours, does not generate the persistence that *regime* and *crisis* usually connote; that would need duration dependence or time-varying transition probabilities [10].

The transition geometry. Figure 3 draws the kernel. Exits from the high state route through the moderate one ($\hat{A}_{32} \approx 0.27$, s.e. 0.030); the direct move to the low state, $\hat{A}_{31}$, is estimated at exactly zero over some 270 entries into the high state. On the boundary of the parameter space this is a bound rather than a point estimate, and all 500 bootstrap replications return zero by construction, being

simulated from, and refitted starting at, a chain without that transition, so the percentile interval is degenerate. With no events in the $n \approx 1,000$ high-state hours the stationary mass implies, the rule of three bounds the probability at $3/n \approx 0.30\%$ per hour. A sample this long could detect that much: simulating 200 samples from the fitted chain with $A_{31} = 0.003$ returns an exact zero in only 6.0% of them (mean estimate 0.0054, 95% interval $[0.000, 0.020]$), and at $A_{31} = 0.010$ never. Volatility steps down through the intermediate level rather than collapsing, consistent with volatility clustering [8] and with Markov-switching GARCH evidence [14]; we know of no earlier report of the property for a decentralised perpetual.

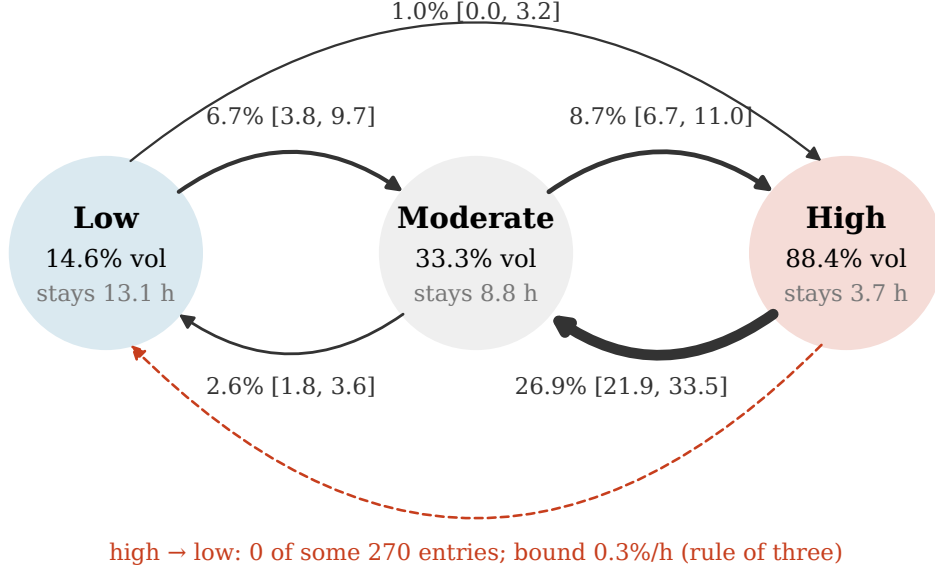


Figure 3. Transition geometry of the $K=3$ return model on the estimation window: hourly transition probabilities with 95% parametric-bootstrap intervals, arrow width increasing with the probability; each node gives the state’s annualised volatility and expected duration. The full-sample fit archived only the diagonal of $\hat{A}$ and the stationary distribution; the off-diagonal probabilities are reconstructed from them with the original fit’s boundary estimate $\hat{A}_{31} = 0$ imposed, so the zero is an input here, and the sub-period fit (Figure 4b) estimates it independently. The bound on $\hat{A}_{31}$ is the rule of three over the roughly 1,000 high-state hours that the stationary mass implies (Section 4.2).

4.3 Robustness to Extreme Observations

Large prints on a leveraged perpetual — liquidation cascades, oracle dislocations, thin-book wicks — are genuine features of the market, so we do not winsorise the estimation sample; the question is whether the high-volatility state rests on a handful of them. It does not. On the hourly sub-period, 3 December 2025 to 28 April 2026 ($T=3,507$), clipping the extreme 0.1% of returns, four observations, lowers the standard deviation from 50.59 to 50.30 bps/h and the refitted high state from 87.7% to 85.4% annualised (Table A1).

The sub-period fit recovers the structure, 12.9%, 31.0% and 87.7% against 14.6%, 33.3% and 88.4%. The high states agree both ways — 87.7% lies inside the full sample’s $[83.0, 94.0]$ and 88.4% inside the sub-period’s $[82.1, 93.7]$ — while the lower two differ by 0.5 to 0.8 points beyond each other’s intervals; $\hat{A}_{31} = 0$ again, and exits from the high state again route through the moderate one, with probability 0.26. Two-hourly candles cover the whole window (2,500 bars, annualised with $\sqrt{12 \cdot 365}$) and give 11.0%, 31.6% and 88.8%, with a high-state mass of 21.0% against 20.0%. Only the boundary result weakens, to $\hat{A}_{31} \approx 0.01$, as it must when a two-hour bar can contain a passage through the moderate state; it stays an hourly statement.

More than three states. On the sub-period, with 40 starts each and both larger models converging, BIC is $-28,356.8$ for $K=2$, $-28,533.4$ for $K=3$, $-28,557.0$ for $K=4$ and $-28,482.2$ for $K=5$, so $K=4$ wins, by 23.6 points; the $K=2$ and $K=3$ figures reproduce the archived sub-sample fit to the second decimal. We keep $K=3$ although $K=4$ is estimable. The gain is small beside the 176.6 points $K=3$ adds to $K=2$; $K=5$ adds only 7.5 log-likelihood points to $K=4$, a pattern a finer discretisation of one

heavy-tailed distribution would also produce; the $K=4$ volatilities, 11.1, 24.8, 55.8 and 134.9 bps/h, form a near-geometric ladder with ratios of 2.2 to 2.4 whose top rung sits where a Gaussian mixture must place mass to reach the tails; and a Student- t GARCH beats the three-state model on BIC with five parameters (Section 4.5). The binding constraint is the Gaussian emission, not the number of volatility levels, and more than three states in one asset have little ex-ante economic interpretation [3].

4.4 Inference

The standard errors of Table 4 come from a parametric bootstrap: 500 series simulated from the fitted chain, state paths started at $\hat{\pi}^\infty$, each refitted with $K=3$ from the generating parameters; against full multi-start refits on 25 paired replications this shortcut costs 0.055 nats on average and moves $\hat{\sigma}$ by 0.014 bps/h. Volatilities and transitions are precise, every $\hat{\sigma}_k$ having a coefficient of variation below 3.5%; the drifts are not.

4.5 Residual Diagnostics and Benchmarks

Conditional independence given the state is testable, and worth testing, since GARCH-type clustering is the competing explanation for a three-state fit. We standardise returns by the one-step predictive moments (4)–(5) from the *filtered* posterior; the smoothed posterior conditions on r_t itself and gives spurious results — $Q(z^2)(10) = 53.3$, $p < 10^{-7}$, with near-Gaussian residuals — shown in Table A2 for contrast. Raw returns on the sub-period cluster overwhelmingly, $Q(z^2)(10) = 470.7$ and $Q(|z|)(10) = 976.0$, both $p < 10^{-90}$. The chain absorbs most of it — $Q(z^2)(10)$ falls to 2.4 and ARCH-LM from 270.3 to 2.4 — but not all: $Q(z^2)$ has little power against ARCH when residuals are as leptokurtic as these, and the tail-robust $Q(|z|)(24) = 96.0$ still rejects. The chain also leaves heavier tails than it started with, an excess kurtosis of 14.4 against 6.9. The MS-AR residuals fail outright ($Q(10) = 42.6$, excess kurtosis 78.9).

Against benchmarks outside its family (Table A3) the three-state HMM beats a single Gaussian and a Gaussian GARCH(1,1) by 1,422 and 831 BIC points but not a Student- t GARCH(1,1), which wins on BIC (−28,583 against −28,533) with five parameters against fourteen while losing on AIC (−28,614 against −28,620). That winner is near-integrated, $\hat{\alpha} + \hat{\beta} = 0.99998$ against the bound of 0.99999 imposed in estimation, so its unconditional variance is barely finite, and with $\hat{\nu} = 3.35$ its innovations have no finite kurtosis: it buys likelihood with tails. Both results point to a Student- t or generalised-hyperbolic emission [11], and the case for the regime model rests on what GARCH does not deliver: a state decomposition, a transition geometry and a state-conditional object that aligns with the funding model.

4.6 Goodness of Fit and Posteriors

Figure 4(a) overlays the stationary mixture $\sum_k \pi_k^\infty \varphi(r; \hat{\mu}_k, \hat{\sigma}_k^2)$ on the sub-period histogram: the mixture tracks the body, and the high state carries almost all the mass beyond $\pm 3\hat{\sigma}_2$, about ± 100 bps/h. Figure 5 follows the posteriors through time. The February 2026 decline to \$62,883 coincides with concentrated high-volatility mass, a move of funding off its floor and a shift of the funding posterior, but high-volatility hours recur throughout the window, as the durations imply. The state-implied volatility $\hat{\sigma}_t^{\text{cond}} = (\sum_k \mathbb{P}(S_t = k | \mathcal{F}_t) \hat{\sigma}_k^2)^{1/2}$, averaged over the trailing 24 hours as realised volatility is, tracks it closely; we have not analysed the lead-lag structure between the two.

5 Regime Estimation: Funding Premium

5.1 Estimated Parameters

Table 5 reports the MS-AR(1) on the paper-window premium. The calm regime has $\hat{\sigma}_1 = 0.53$ bps and $\hat{\phi}_1 = 0.981$; the stressed regime has $\hat{\sigma}_2 = 2.68$ bps, 5.1 times larger, and $\hat{\phi}_2 = 0.804$. The levels, $\hat{\mu}_1 = -2.78$ and $\hat{\mu}_2 = -3.22$ bps, sit beside the sample mean: the regimes differ in volatility and persistence, not in level. Near the unit root the half-life is steep in ϕ — $d\tau/d\phi \approx 1,860$ hours per unit at 0.981 — so even with a standard error of 0.0037 the calm half-life is known only to within a factor of two, 22 to 49 hours. $\hat{\phi}_1$ still separates from unity: its t -ratio of −5.3 clears the Dickey–Fuller 1% value of about −3.4 that a test on the stationarity boundary faces.

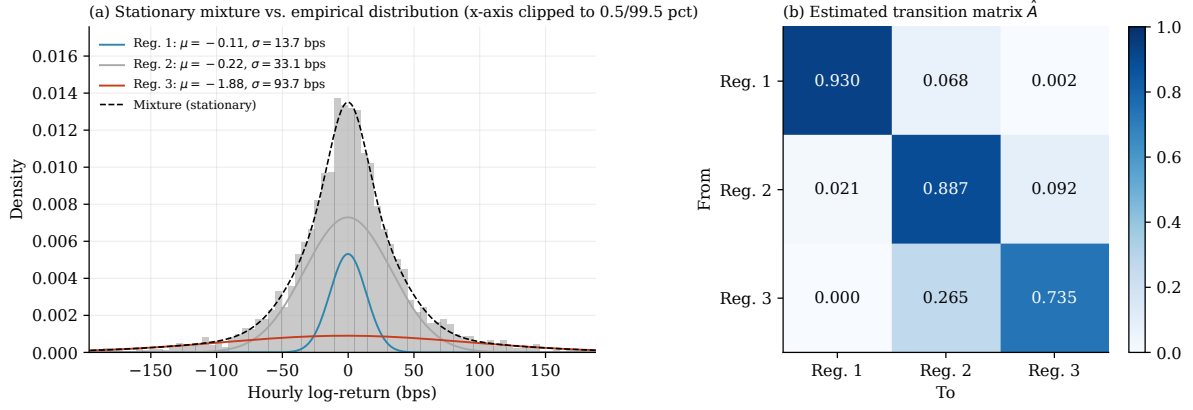


Figure 4. $K=3$ Gaussian HMM on hourly log-returns, fitted to the hourly sub-period (3 Dec 2025 – 28 Apr 2026, $T = 3,507$), the window on which hourly returns survive. **(a)** Empirical density (grey) against the stationary mixture, components weighted by π^∞ ; the horizontal range is truncated at the 0.5th and 99.5th percentiles, so the panel speaks for the body of the distribution, not its tails. **(b)** Estimated kernel $\hat{A}$; the zero at (3, 1) is the high-to-low transition of Section 4.2. Reg. k is State k of Table 4.

Table 5. MS-AR(1) parameter estimates for the funding premium y_t (8h-equivalent bps). $\hat{\mu}_k$ is the regime *level* in (2), not an intercept; half-life follows (3). Parenthesised figures are standard errors of $\hat{\phi}_k$, and the intervals are delta-method; Section 7 explains why both are indicative rather than exact.

Regime	$\hat{\mu}_k$ (bps)	$\hat{\phi}_k$	$\hat{\sigma}_k$ (bps)	Half-life (h)	95% CI
1 (calm)	-2.78	0.9807 (0.0037)	0.525	35.6	[22.2, 48.9]
2 (stressed)	-3.22	0.8040 (0.0462)	2.681	3.2	[1.5, 4.8]
Ratio $\hat{\tau}_1/\hat{\tau}_2 = 11.2$ (s.e. 4.2; 95% CI [3.0, 19.4], delta method on the full covariance)					
$\log \mathcal{L} = -4808.53$, AIC = 9633.1, BIC = 9685.2, $T_{\text{eff}} = 5,000$					

5.2 The Mean-Reversion Asymmetry

The half-lives are about 36 hours calm and 3.2 hours under stress, a ratio of 11.2 with a delta-method 95% interval of [3.0, 19.4] (Figure 6): on this window the premium reverts an order of magnitude faster under stress. A size-of-prize mechanism would give this sign. A calm dislocation of a few basis points may not cover the fees, margin and inventory risk of the offsetting basis trade, so capital is not deployed and the dislocation decays slowly; a stressed one makes funding a first-order part of position PnL, clears those costs and is arbitrated back within hours. The argument concerns the size of the mispricing, not liquidity, which on its own would predict the opposite ordering. The specification is a state-dependent Ornstein–Uhlenbeck process. We know of no comparable estimate for a decentralised perpetual, but ours is a single-venue, single-asset point estimate and we claim no priority on its magnitude. Section 5.3 shows that the ordering belongs to one episode.

5.3 Specification Robustness and the Precision of the Asymmetry

Two optima, and the frequent one is wrong. The likelihood of (2) has two well-separated maxima on this series, and each of our 240 estimations reaches one of them (Figure 7). The inferior one lies 20.1 points lower, a likelihood ratio of order 10^8 , yet attracts most estimations: 24 of the paper’s 40, and across six disjoint blocks of 40 seeds the inferior mode attracts between 20 and 30 of the 40 estimations, while the ranking of the two modes never changes. It differs from the maximum by 0.13 in $\hat{\phi}_2$ (0.672 against 0.804) and 0.003 in $\hat{\phi}_1$, which the steep half-life turns into a ratio of 24.6 instead of the 11.2 of Table 5. `statsmodels` runs no random-start search unless asked, and even with a 40-draw search a single estimation lands on the inferior optimum three times in five: a Markov-switching result resting on a nonlinear function of $\hat{\phi}$ needs a recorded multi-start protocol and a reported terminal log-likelihood. We set $K=2$ on interpretability and on the small number of stress episodes the window contains.

BTC perpetual on Hyperliquid — return regimes (K=3) and funding regimes

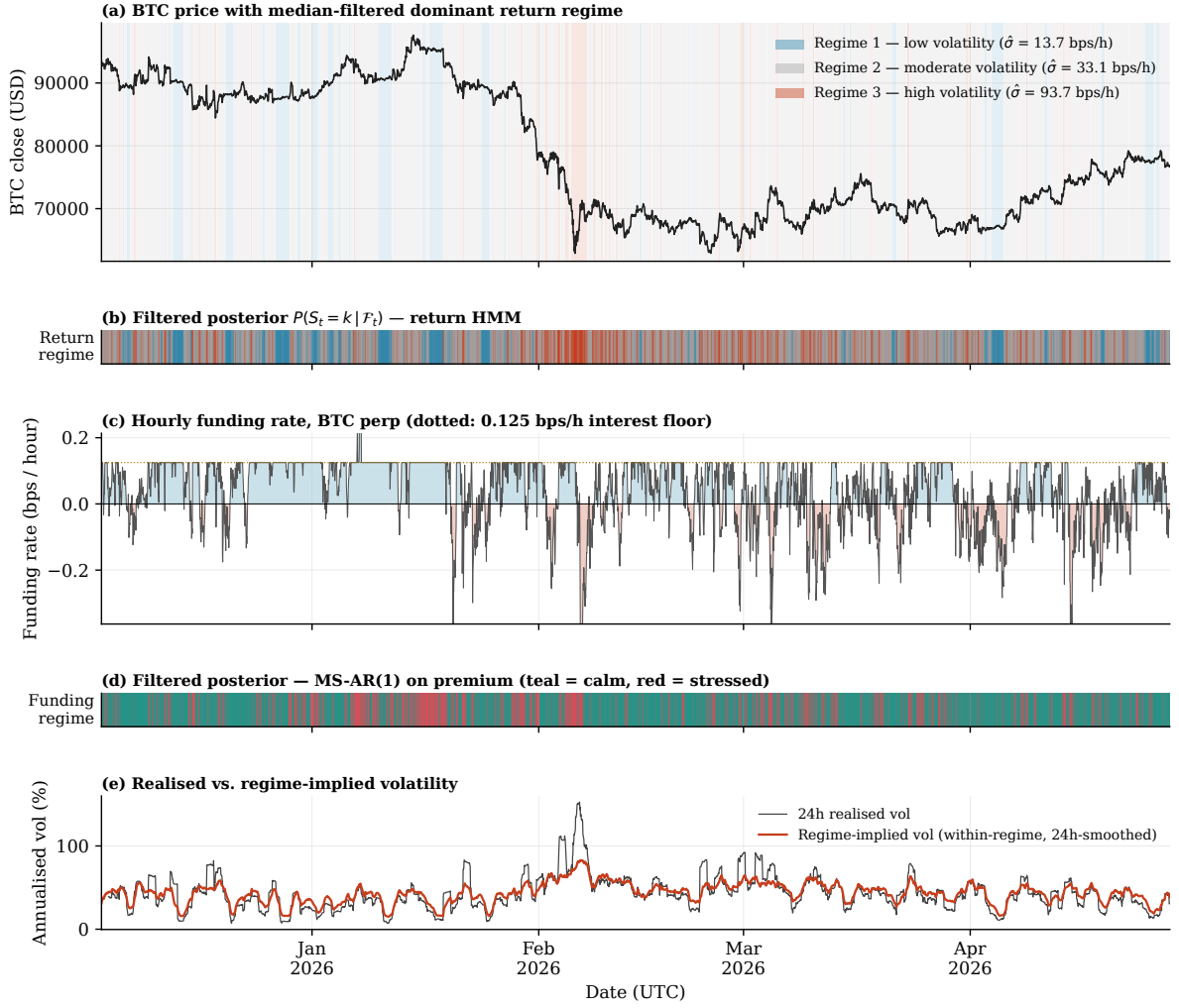


Figure 5. State-detection dashboard over the hourly sub-period, 3 Dec 2025 – 28 Apr 2026 ($T = 3,507$). **(a)** Closing price, shaded by the dominant return state after a centred seven-hour median filter, which is non-causal and presentational only; panel (b) is the causal object. **(b)** Filtered posterior $\mathbb{P}(S_t = k | \mathcal{F}_t)$ of the $K=3$ HMM as mixed colour, blue, grey and red for the low-, moderate- and high-volatility states. **(c)** Hourly funding rate; the horizontal segments at 0.125 bps/h are the floor of Section 3.2, and the vertical range runs from the 0.5th to the 99.5th percentile with a margin and headroom above the floor, so a few extreme prints fall outside it. **(d)** Filtered posterior of the MS-AR(1) on the premium re-estimated on this window, where the half-life ordering is reversed (Section 5.3); it is not the fit of Table 5. **(e)** 24-hour realised volatility against the 24-hour trailing mean of the state-implied conditional volatility, both annualised; the latter is a nowcast, not a forecast.

Where the asymmetry lives. The premium series can be refitted on any window (Figure 8b). On the 3 December 2025 – 28 April 2026 sub-period the ordering reverses, 7.1 hours calm against 24.8 stressed, at 0.29 with a 95% interval of $[0.05, 0.52]$; 36 of 40 estimations reach that optimum, the other four fail to converge, and none finds a better one. On the whole series, 2 October 2025 to 30 September 2026 ($T = 8,737$), the original ordering returns at 4.55, $[1.95, 7.15]$. The sub-period contains the February 2026 episode entire, including the premium minimum of -10.55 bps on 6 February; it lacks the maximum, $+23.61$ bps in one hour of the 10 October 2025 crash (Figure 8a), inside the two-hour bar that sets the 1,498 bps bound of Table 1. The stressed state is identified by that hour, not by the February selloff: a window excluding October does not identify the state, and one diluting it attenuates the estimate. Nor do $[3.0, 19.4]$ and $[1.95, 7.15]$ bracket the ratio: the window from the end of the estimation sample to 30 September 2026 gives an interval ending at 0.49, disjoint from both and on the far side of unity.

Fourteen overlapping windows of 2,500 hours, stepped 450, localise the effect (Figure 9, Table A4). Only the first, the one containing 10 October 2025, places its interval above unity, at 39.17 with $[7.6, 70.8]$; two others have point estimates above one, 2.94 and 1.11, without separating from it, and eleven of the

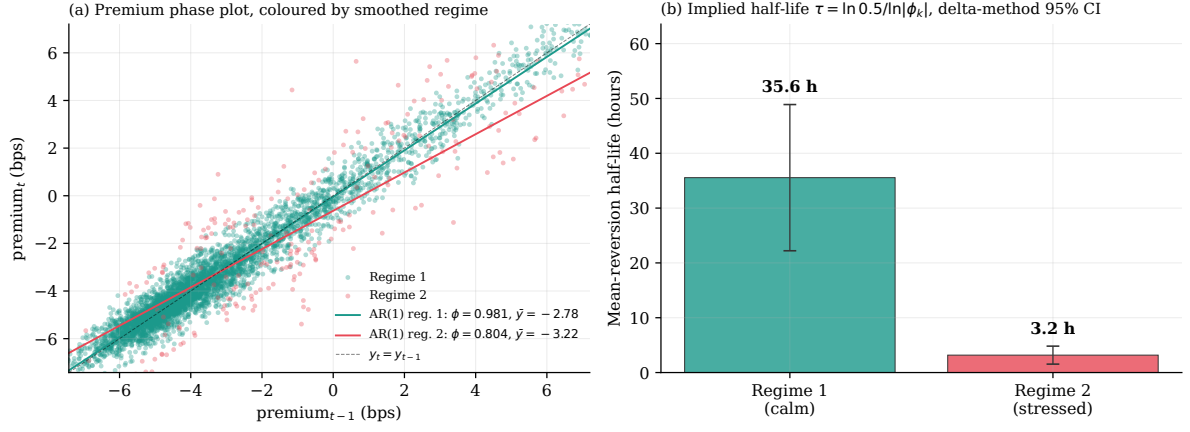


Figure 6. Funding-premium dynamics under the MS-AR(1) of Table 5. **(a)** Phase plot of (y_{t-1}, y_t) in bps, coloured by the modal *smoothed* regime, an in-sample diagnostic only; the lines are the regime-conditional fits $\hat{\mu}_k(1 - \hat{\phi}_k) + \hat{\phi}_k y_{t-1}$. Both axes are truncated at the 0.5th and 99.5th percentiles, so the October 2025 excursion to +23.61 bps that identifies the stressed state falls outside the range. **(b)** Half-life per regime with delta-method intervals.

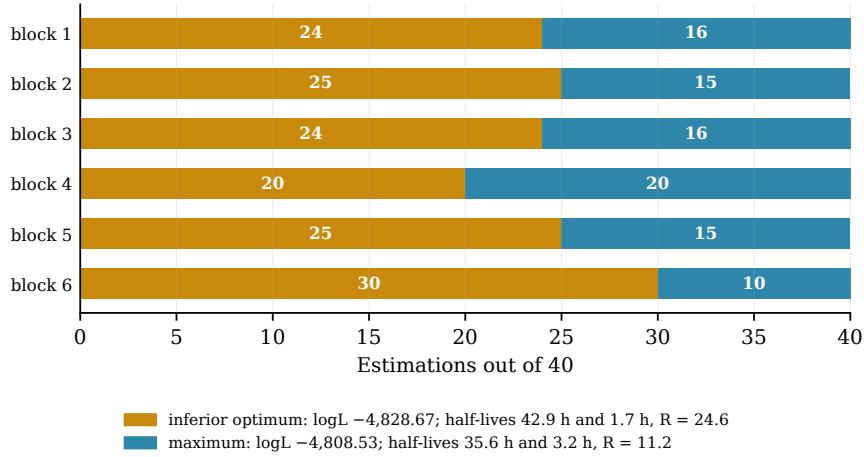


Figure 7. The two optima of the premium likelihood on the paper window. Each bar is a block of 40 consecutive seeds (the first starts at seed 20260802), split by the optimum each estimation reaches; the legend gives what each optimum implies.

fourteen fall below one outright and nine of those exclude unity. The estimate falls as October leaves the window, crosses unity at the turn of the year and never returns, while the mean drawdown deepens from 21 to 47 per cent: one episode, then a baseline. Overlapping by 82%, they are fourteen views of a series holding three disjoint windows of this length, not fourteen tests, and twelve months with one drawdown support no cyclical reading.

One observation carries the significance. The 10 October hour is a genuine observation, not a recording error, and it stays in every main estimate, Table 5 included. We replace it only to measure what it carries: a two-state model can devote its high-variance state to a single extreme hour, and this one is two and a half times the next largest. Replacing it with that next largest value, 9.52 bps, moves the paper-window ratio from 11.19 to 3.04 and deleting it to 1.83, both on intervals that cover unity, because $\hat{\tau}_{\text{stressed}}$ moves from 3.18 hours to 11.15 and 18.12 while the calm half-life barely shifts (Figure 8b, Table A5). Winsorising the upper 1 per cent, fifty hours, still leaves 4.49 with an interval above unity: the tail carries part of the effect, but the exclusion of unity rests on one hour in 5,001.

The converse is the sharper test. On the window from the end of the estimation sample to 30 September 2026 the ratio is 0.226, 11.0 hours calm against 48.7 stressed, its 95% interval ending at 0.49 with the lower end left open by the delta method: there the stressed state is the persistent one. Overwriting a single hour with a synthetic +23.61 bps, at each quartile of that window in turn, raises it to 18.0, 11.4 and 4.3 — a factor of 19 to 79 on one observation in 3,736. At the first point the

manufactured ratio is even significant, [1.97, 33.98]; the other two delta-method intervals cover unity. We report these as fragility, not as estimates: one hour can manufacture an asymmetry of the headline's order, significance included, and where it falls decides how large.

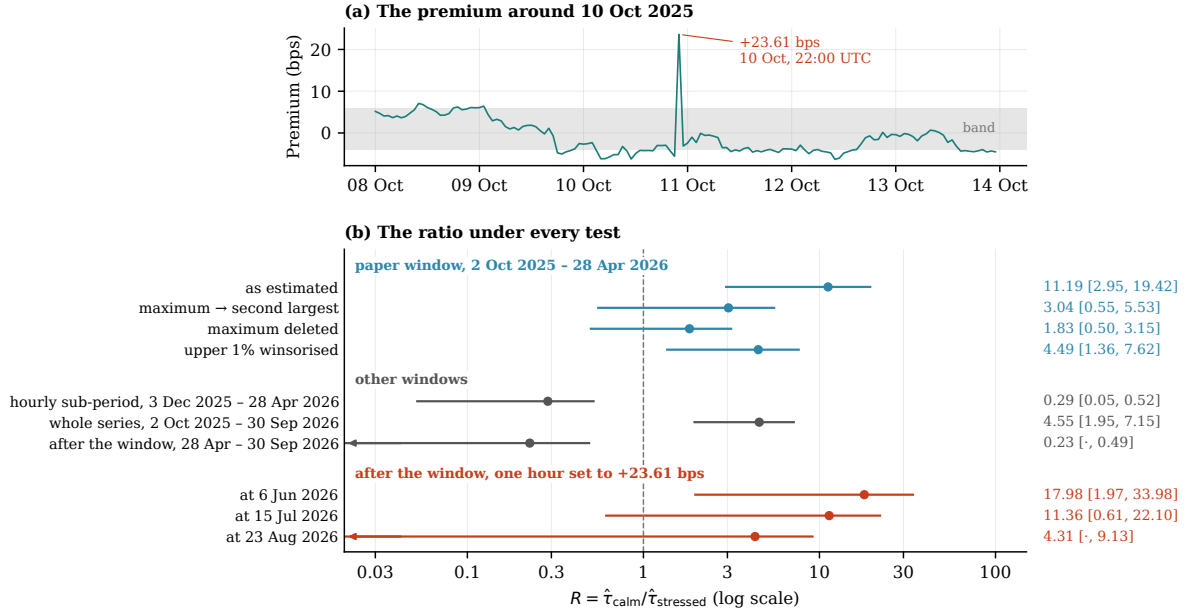


Figure 8. The hour that carries the asymmetry. **(a)** Hourly premium, 8–13 October 2025, against the band; the hour of 10 October 22:00 UTC is the maximum of the whole series. **(b)** The half-life ratio with its 95% delta-method interval on the paper window and its upper-tail variants (Table A5), on the other windows, and on the later window with one hour overwritten at each quartile; a lower end the delta method returns negative is left open (arrow).

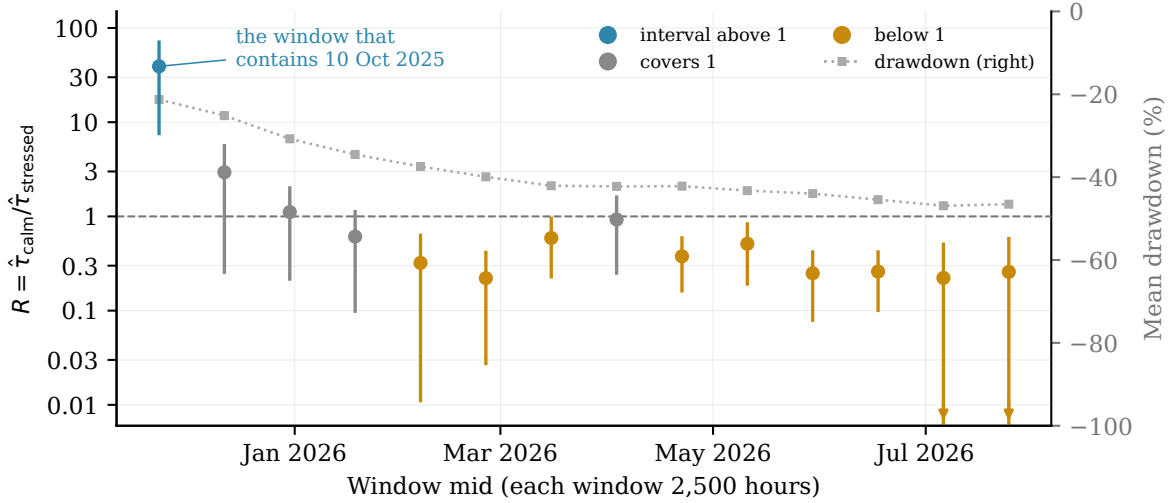


Figure 9. Rolling estimates of the half-life ratio, windows of 2,500 hours stepped by 450, at their mid-points: blue if the 95% interval lies above unity, grey if it covers unity, ochre if it lies below; arrows mark lower ends the delta method leaves open. The dotted line with squares is the mean drawdown from the running price peak (right axis). Table A4 lists the estimates.

What the later data replicate, and what we claim. Over the 3,736 hours after the estimation window, 51.7% of funding observations sit exactly on the interest floor, against 46.0% inside it: the censoring persists, if anything more often. The premium's two-state structure does not: the ratio of conditional standard deviations across states falls from 5.1 to 2.1, the reversed half-life ordering seen through the variance rather than the persistence. The asymmetry is therefore a property of the October 2025 episode, not a parameter of the Hyperliquid BTC funding premium. Table 5 describes that episode,

and the size-of-prize mechanism remains a candidate explanation for it, which only several such episodes could establish.

6 Implications

Neither implication is tested (Section 7). *Position sizing.* The filtered posterior in (4)–(5) gives a one-step conditional variance that a mean-variance rule $w_t = \mathbb{E}[r_{t+1} | \mathcal{F}_t] / (\lambda \text{Var}(r_{t+1} | \mathcal{F}_t))$ would turn into time-varying positions without a separate volatility target. Its within-state term spans 14.6% to 88.4% annualised, a 6.1-fold range; its between-state term never exceeds 0.3% of the forecast variance, the state means spanning only 6.1 bps/h, so regime uncertainty as such does not move positions for this asset, though it would for one with a larger spread of state-conditional drifts. *Funding-aware execution.* Calm-regime funding is a slow carry cost and stressed-regime funding a transient, but the only available figure rests on $\hat{\tau}_{\text{stressed}} = 3.2$ hours, which becomes 11.15 or 18.12 hours if one observation changes (Table A5) and is 49 hours on the window after April 2026 (Section 5.3). Whether an execution schedule could exploit the October pattern is a backtest’s question, and we do not answer it.

7 Limitations

Scope. Every result is in-sample and descriptive. Posteriors are evaluated at the full-sample estimate, the filtered/smoothed distinction is a property of the recursions rather than a real-time claim, and no walk-forward, backtest or transaction-cost exercise is run.

External validity. One asset on one venue: seven months of returns with a single macro drawdown and twelve months of premium containing a single funding episode, compared with two exchanges in one market phase, in which their premiums sat below the band; with persistently positive premia the ordering could change. We would not carry the parameter values to other assets, venues or periods.

Inference. The HMM bootstrap conditions on the Gaussian emission, which the diagnostics reject; a block bootstrap would relax that, and the heteroskedasticity left after the state decomposition makes a Markov-switching GARCH [4, 14] the natural next model. The MS-AR standard errors come from the Hessian of a Gaussian likelihood its residuals reject badly — excess kurtosis of 79, a Jarque–Bera statistic more than forty times the HMM’s — so its delta-method intervals are indicative; a sandwich or block-bootstrap covariance would be the repair, and we have not computed one. The ordering of the two point estimates, 35.6 hours against 3.2, does not involve the covariance, but $\hat{\phi}_{\text{stressed}}$ is itself set by one observation.

Specification. The three-state HMM does not dominate a Student- t GARCH(1,1) on BIC, and we run no encompassing test. No third premium state is tested, the nuisance-parameter problem of Section 2.1 applying here too [9, 16]. The premium index is a within-hour average, which induces a moving-average component, so the AR(1) is misspecified and $\hat{\phi}$ biased; a Markov-switching ARMA(1,1) is the next step.

Replication. Every funding-side result reproduces from source; the hourly return panel’s first two months do not (Data and Code Availability).

8 Conclusion

The measurement we would most like carried forward is the censoring. One rule, common to the major perpetual venues, pins funding to the interest rate in 46% of Hyperliquid’s settlements and in only 6 to 8% of Binance’s and Bybit’s over the same window, because the exchanges’ premiums sit just below the band while Hyperliquid’s straddles its edge. Work on Hyperliquid funding should model the premium, and cross-venue comparisons that read funding as positioning should compare premiums, with their windows and indices in view. Why the premiums sit where they do — index construction, the depth of arbitrage capital, the mix of traders — is the question the measurement opens, and it needs more venues, assets and market phases.

On returns, the estimated probability that hourly volatility moves directly from the high state to the low is zero, with a 95% bound of 0.3% per hour. On the premium, the asymmetry in mean reversion describes the October 2025 episode and not the series. The natural extensions are a Student- t or generalised-hyperbolic emission [11], time-varying transition probabilities driven by covariates such as open interest or liquidity [10], and a regime-conditional carry backtest once the asymmetry has been seen across several episodes.

Data and Code Availability

The estimation panel comes from Hyperliquid’s public REST API — `candleSnapshot` (hourly OHLCV candles of traded prices) and `fundingHistory` (hourly funding rate and premium index) — retrieved on 28 April 2026; the premium series beyond it and the two-hourly candles were retrieved from the same endpoints in August and October 2026. Binance and Bybit data come from their public endpoints (Binance: `fapi/v1/fundingRate` and `fapi/v1/premiumIndexKlines`; Bybit: `v5/market/funding/history` and `v5/market/premium-index-price-kline`), retrieved on 6 October 2026. All endpoints are unauthenticated, and no proprietary or licensed data are used.

The *funding* and *premium* series is fully replicable: `fundingHistory` has no retention limit, and an independent re-retrieval agreed with the archived panel to machine precision on all 3,507 overlapping hours. The hourly *price* series is not: the results that use hourly prices before 3 December 2025 — the log-return and closing-price rows of Table 1, Tables 3–4, Figure 3, which draws the full-sample kernel, and the price labels of Figure 1 — rest on the original extract. Every other figure is drawn from the shipped data and artifacts: Figures 4 and 5 on the archived hourly panel of 3 December 2025 – 28 April 2026, and Figure 6 on the paper window of the premium series. The materials also contain two-hourly candles for the whole span (Figure 1); the exchange’s two-hourly window will cease to cover 2 October 2025 in late November 2026, its four-hourly window in 2028.

Code, data, the fitted parameters behind every table, figure and quoted fit in machine-readable form — except the full-sample $K=2$ fit, whose log-likelihood survives only in Table 3 — and the seeds and multi-start protocol behind each estimate are in the replication repository,

<https://github.com/gelatotrade/HMM-Decentralised-Perpetual-Futures>.

Competing Interests

The author does not trade on Hyperliquid, Binance or Bybit, holds no equity, advisory or employment relationship with any of them, and received no compensation connected with this work; no third party reviewed or influenced the analysis. The Hyperliquid Research Collective, which publishes this research under the author’s name alongside this posting, is an independent group despite the shared name and has no relationship to Hyperliquid; the author’s work for it is unpaid.

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A Additional Tables

Table A1. Robustness of the $K=3$ state volatilities to winsorisation, estimated on the reproducible sub-period (3 Dec 2025 – 28 Apr 2026, $T = 3,507$). Winsorisation clips the most extreme 0.1% of hourly returns, four observations in total. Under the $\hat{\sigma}$ -ascending labelling of Section 2.1 index and state coincide, so column k is State k of Table 4. The estimated probability of a direct high-to-low-volatility transition remains zero in both fits, and the stationary masses move by at most 1.2 percentage points (high-volatility state 21.2% $\rightarrow$ 22.4%, low-volatility state 18.3% $\rightarrow$ 17.3%).

Annualised $\hat{\sigma}$	1 low	2 moderate	3 high
Raw ($T = 3,507$)	12.9%	31.0%	87.7%
Winsorised (extreme 0.1%)	12.6%	30.4%	85.4%

Table A2. Residual diagnostics. Raw-return and HMM rows use the sub-period panel ($T = 3,507$); MS-AR rows use the paper window ($T = 5,001$, leaving 4,999 residuals — the autoregressive lag costs one observation and the state-pair construction of the one-step forecast another), because that window’s premium series survives and its return series does not. For the HMM rows z_t is standardised by the one-step-ahead mixture mean and the full conditional variance of (5); headline rows use filtered (causal) state probabilities, smoothed rows are shown for contrast only.

<i>Panel A: Ljung–Box $Q(\ell)$, p-value in brackets</i>				
Series	ℓ	$Q(z)$	$Q(z^2)$	$Q(z)$
Raw log-returns	10	6.7 [0.76]	470.7 [<0.0001]	976.0 [<0.0001]
	24	17.0 [0.85]	671.2 [<0.0001]	1,718.1 [<0.0001]
HMM $K=3$, filtered	10	7.9 [0.64]	2.4 [0.99]	20.6 [0.024]
	24	18.3 [0.79]	5.7 [1.00]	96.0 [<0.0001]
HMM $K=3$, smoothed	10	14.2 [0.16]	53.3 [<0.0001]	32.8 [0.0003]
	24	28.2 [0.25]	92.3 [<0.0001]	78.4 [<0.0001]
MS-AR(1), paper window	1	6.7 [0.0096]	48.9 [<0.0001]	36.8 [<0.0001]
	10	42.6 [<0.0001]	51.5 [<0.0001]	90.3 [<0.0001]
<i>Panel B: heteroskedasticity and normality</i>				
Series	ARCH-LM(12)	Jarque–Bera	Skew	Exc. kurt.
Raw log-returns	270.3 [<0.0001]	7,081 [<0.0001]	−0.26	6.94
HMM $K=3$, filtered	2.4 [0.9985]	31,101 [<0.0001]	−1.02	14.45
HMM $K=3$, smoothed	46.1 [<0.0001]	13 [0.0017]	−0.06	0.27
MS-AR(1), paper window	51.4 [<0.0001]	1,304,072 [<0.0001]	2.80	78.93

Table A3. Out-of-family comparison on the sub-period panel ($T = 3,507$), all rows on the same raw log-return series with $AIC = 2k - 2\log \mathcal{L}$ and $BIC = k\log T - 2\log \mathcal{L}$; lower is better. Log-likelihoods are not scale-free, so mixing return units across rows would decide the ranking by itself. The Student- t GARCH row, which wins on BIC, is near-integrated ($\hat{\alpha} + \hat{\beta} = 0.99998$, against the bound of 0.99999 imposed in estimation) with $\hat{\nu} = 3.35$: a barely finite unconditional variance, no finite kurtosis. Both qualifications are recorded in `output/benchmarks.json`.

Model	$\log \mathcal{L}$	# params	AIC	BIC
GARCH(1,1)-Student- t	14,312.1	5	−28,614.2	−28,583.4
Gaussian HMM, $K=3$	14,323.8	14	−28,619.7	−28,533.4
Gaussian HMM, $K=2$	14,207.0	7	−28,400.0	−28,356.8
GARCH(1,1)-Normal	13,867.3	4	−27,726.7	−27,702.0
Gaussian i.i.d.	13,563.8	2	−27,123.5	−27,111.2

Table A4. Rolling MS-AR(1) estimates of $R = \hat{\tau}_{\text{calm}}/\hat{\tau}_{\text{stressed}}$ on the premium series, windows of 2,500 hours stepped by 450, forty starts each; the final 387 hours of the series fill no further window. *Side* records which side of unity the delta-method interval falls on; $\cdot$ marks a lower endpoint the delta method returns negative, which a half-life cannot be. *Price* is the mean two-hourly closing price over the window and *DD* its mean drawdown from the running peak. Regimes are labelled by $\hat{\sigma}_k^2$ ascending throughout.

Window mid	$\hat{\tau}_{\text{calm}}$	$\hat{\tau}_{\text{str}}$	$\hat{R}$	95% CI	side	price	DD
2025-11-23	50.91	1.30	39.17	[7.59, 70.75]	> 1	98,691	−21%
2025-12-11	37.71	12.81	2.94	[0.26, 5.63]	n.s.	93,944	−25%
2025-12-30	18.26	16.42	1.11	[0.22, 2.01]	n.s.	86,913	−31%
2026-01-18	10.98	17.96	0.61	[0.10, 1.12]	n.s.	82,165	−35%
2026-02-06	7.59	23.63	0.32	[0.01, 0.63]	< 1	78,522	−37%
2026-02-24	6.18	27.93	0.22	[0.03, 0.41]	< 1	75,393	−40%
2026-03-15	4.88	8.26	0.59	[0.23, 0.95]	< 1	72,701	−42%
2026-04-03	5.79	6.24	0.93	[0.25, 1.61]	n.s.	72,487	−42%
2026-04-22	3.41	9.05	0.38	[0.16, 0.59]	< 1	72,581	−42%
2026-05-10	3.92	7.64	0.51	[0.19, 0.83]	< 1	71,215	−43%
2026-05-29	4.93	19.78	0.25	[0.08, 0.42]	< 1	70,321	−44%
2026-06-17	4.39	16.87	0.26	[0.10, 0.42]	< 1	68,481	−45%
2026-07-06	9.36	42.04	0.22	[$\cdot$, 0.51]	< 1	66,633	−47%
2026-07-24	10.11	39.22	0.26	[$\cdot$, 0.58]	< 1	67,110	−47%

Table A5. Sensitivity of the paper-window asymmetry to the upper tail of the premium. Each row refits (2) on 2 October 2025 – 28 April 2026 after one change, with the baseline refitted alongside under the same protocol. *Maximum* $\rightarrow$ *2nd largest* replaces the +23.61 bps observation of 10 October 2025 by 9.52 bps; *Maximum deleted* removes it.

Series	T	$\hat{\tau}_{\text{calm}}$	$\hat{\tau}_{\text{str}}$	$\hat{R}$	95% CI	side
Unchanged	5,001	35.55	3.18	11.19	[2.95, 19.42]	> 1
Maximum $\rightarrow$ 2nd largest	5,001	33.90	11.15	3.04	[0.55, 5.53]	n.s.
Maximum deleted	5,000	33.10	18.12	1.83	[0.50, 3.15]	n.s.
Winsorised, upper 1%	5,001	42.23	9.40	4.49	[1.36, 7.62]	> 1

B Estimation Details

Recursions. The filtered posterior is computed through the forward variable $\alpha_t(i) = \mathbb{P}(r_{1:t}, S_t = i \mid \theta)$,

$$\alpha_1(i) = \pi_i b_i(r_1), \quad \alpha_{t+1}(j) = \left[\sum_i \alpha_t(i) A_{ij} \right] b_j(r_{t+1}), \quad (\text{B1})$$

and the smoothed posterior from the forward and backward variables,

$$\gamma_t(i) = \frac{\alpha_t(i) \beta_t(i)}{\sum_j \alpha_T(j)}, \quad \beta_t(i) = \sum_j A_{ij} b_j(r_{t+1}) \beta_{t+1}(j); \quad (\text{B2})$$

the Viterbi path replaces the marginalisation in (B1) with maximisation. Baum–Welch alternates an E-step for γ and the pairwise posteriors $\xi_t(i, j) = \mathbb{P}(S_t = i, S_{t+1} = j \mid \mathcal{F}_T, \theta^{(n)})$ with the closed-form M-step

$$\hat{A}_{ij} = \frac{\sum_t \xi_t(i, j)}{\sum_t \gamma_t(i)}, \quad \hat{\mu}_k = \frac{\sum_t \gamma_t(k) r_t}{\sum_t \gamma_t(k)}, \quad \hat{\sigma}_k^2 = \frac{\sum_t \gamma_t(k) (r_t - \hat{\mu}_k)^2}{\sum_t \gamma_t(k)}. \quad (\text{B3})$$

EM finds local maxima, so we run eight random initialisations and keep the highest terminal log-likelihood. At $T \sim 5,000$ the recursions underflow in linear space; we run them in log space with the *logsumexp* trick, vectorised across destination states. The implementation is self-contained and validated against simulated ground truth.

Labelling. Label switching [30] requires an ordering rule, and the natural one for returns, $\hat{\mu}_1 \leq \dots \leq \hat{\mu}_K$, fails here: two of the three means lie within 0.27 bps/h of each other, well inside their sampling variability. In the parametric bootstrap of Section 4.4 the μ - and σ -ascending orderings disagree in 40.2% of replications, and under μ -ascending the 95% interval for the state with $\hat{\sigma} = 15.6$ bps/h is $[14.6, 90.7]$, against $[14.5, 16.5]$ under σ -ascending: the first interval measures the sorting rule, not the estimator. A μ -ascending rule would also permute the upper two indices of the full-sample and sub-period fits relative to each other, so that the transition regularity of Section 4.2 would seem not to replicate. The volatilities are separated by many standard errors, and the σ -ascending order is stable in every fit we report.

The premium equation. In (2) μ_k is the regime level; the fixed-coefficient form would carry the intercept $c_k = \mu_k(1 - \phi_k)$, and with $\hat{\phi}$ near unity $\hat{c}_k/(1 - \hat{\phi}_k)$ is badly behaved, so we report $\hat{\mu}_k$ throughout. The `statsmodels` estimator combines an EM initialisation, a random start search and quasi-Newton optimisation.